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Raj Kumar Paudel

Ph.D., Postdoctoral Research Fellow

Summary

I am a computational physicist working on the electronic structure, excitons, and quantum transport of two-dimensional materials. My published work covers the band structures and optical properties of graphene, graphene nanoribbons, and transition-metal dichalcogenides, including interlayer excitons in bilayer MoS_2 and electric-field-driven hole transfer between the layers of bilayer WSe_2. I wrote the semi-empirical pseudopotential codes behind much of this work myself, in Fortran and Python, and I have used group theory in practice — our density functional theory scheme for fullerenes is built on a symmetrized icosahedral basis. I now work on quantum transport, developing non-equilibrium Green's function code for F_4TCNQ-doped graphene, a system in which the adsorbed molecule withdraws charge from the surface.

Education

Sep 2016 – Jul 2023 **Ph.D. in Physics**, *Taiwan International Graduate Program (TIGP), Academia Sinica & National Central University*, Taipei, Taiwan
Dissertation: Semi-Empirical Pseudopotential Method for Low-Dimensional Materials
Supervisor: Prof. Yia-Chung Chang

Oct 2006 – Jun 2009 **M.Sc. in Physics**, *Central Department of Physics, Tribhuvan University*, Kirtipur, Nepal

Oct 2002 – Sep 2006 **B.Sc. in Physics (Major)**, *Tribhuvan University*, Kathmandu, Nepal

Research Experience

Aug 2025 – Present **Postdoctoral Research Fellow**, *Department of Physics, National Cheng Kung University (NCKU)*, Tainan, Taiwan
- Developing non-equilibrium Green's function (NEGF) quantum-transport methods for mesoscopic and low-dimensional devices.
- Modeling charge transfer and transport in molecule-functionalized 2D materials, including F_4TCNQ-doped graphene (molecule–material interface).
- Running large-scale simulations for nanoelectronic and quantum-material systems on HPC resources.

Supervisor: Prof. Ming-Hao Liu

Aug 2023 – Jul 2025 **Postdoctoral Research Fellow**, *Research Center for Applied Sciences (RCAS), Academia Sinica*, Taipei, Taiwan
- Developed and applied SEPM and DFT methods to the electronic and optical properties of TMDCs, graphene, and fullerene systems.
- Modeled exciton binding energies, interlayer excitons, and optical response in monolayer and bilayer TMDCs.
- Investigated interlayer and molecular charge-transfer mechanisms in low-dimensional and C_{60} systems.

Supervisor: Prof. Yia-Chung Chang

Sep 2016 – Jul 2023 **Ph.D. Researcher**, *Academia Sinica & National Central University (TIGP)*, Taipei, Taiwan
- Developed and implemented the Semi-Empirical Pseudopotential Method (SEPM) for low-dimensional materials.
- Applied SEPM to the electronic structure of graphene, armchair graphene nanoribbons, and monolayer TMDCs.
- Parameterized local and nonlocal pseudopotentials for multiple material systems.

Supervisor: Prof. Yia-Chung Chang

Teaching Experience

Aug 2014 – Jul 2016 **Physics Lecturer (Higher Secondary)**, *Motherland School*, Masbar, Pokhara, Nepal
- Taught Grade 11–12 physics — optics, electromagnetism and modern physics — under the HSEB national curriculum.
- Supervised the physics practical laboratory and prepared students for the HSEB board examinations.

Sep 2010 – Aug 2013 **Physics Lecturer (part-time, contract)**, *Kathmandu Engineering College, Kalimati*, Kathmandu, Nepal
- Taught undergraduate engineering physics and ran the accompanying physics laboratory.
- Held concurrently, on a contract basis, with the position at Trinity International College.

Sep 2009 – Jul 2014 **Physics Lecturer (Higher Secondary)**, *Trinity International College*, Dillibazar, Kathmandu, Nepal
- Taught Grade 11–12 physics — mechanics, electromagnetism and modern physics — under the HSEB national curriculum.
- Supervised the physics practical laboratory and prepared students for the HSEB board examinations.

Technical Skills

Methods	Density Functional Theory (DFT); Non-equilibrium Green's Function (NEGF) quantum transport; Semi-Empirical Pseudopotential Method (SEPM); exciton physics and optical-response modeling; symmetry and group-theory analysis (irreducible representations).
Programming	Fortran, Python, C++, Bash; NumPy, SciPy, Matplotlib; Git/GitHub; parallel/HPC computing.
Software	Quantum ESPRESSO, SIESTA, VASP, Wannier90, VESTA, XCrySDen.
Material Systems	Graphene, graphene nanoribbons, TMDCs, fullerenes, van der Waals heterostructures, bilayer systems, organic-molecule-doped graphene.
Languages	Nepali (native), English (fluent), Chinese (basic).

Awards and Honors

- Taiwan International Graduate Program (TIGP) full scholarship for Ph.D. studies, Academia Sinica (2016–2023)

Publications

Complete publication list available at Google Scholar.

Peer-Reviewed Journal Articles

- Cha, S., Xu, Z., Ouyang, T., Yao, H., **Paudel, R. K.**, Taniguchi, T., Watanabe, K., Joe, A. Y., Chang, Y.-C., Gabor, N. M., & Lui, C. H. Hybridization-revealed interlayer 2p exciton state in bilayer MoS_2. *Nature Physics*, accepted (2026).
- **Paudel, R. K.**, Ren, C.-Y., & Chang, Y.-C. Semi-empirical Pseudopotential Method for Monolayer Transition Metal Dichalcogenides. *Computer Physics Communications* **327**, 110240 (2026). DOI: 10.1016/j.cpc.2026.110240
- Ouyang, T., Liu, E., Cha, S., Cang, R., **Paudel, R. K.**, Sun, Y., Xu, Z., Taniguchi, T., Watanabe, K., Gabor, N. M., Chang, Y.-C., & Lui, C. H. Brightening Interlayer Excitons by Electric-Field-Driven Hole Transfer in Bilayer WSe_2. *Physical Review Letters* **136**, 096903 (2026). *Editors' Suggestion.* DOI: 10.1103/mjhj-83wc
- **Paudel, R. K.**, Ren, C.-Y., & Chang, Y.-C. Semi-Empirical Pseudopotential Method for Graphene and Graphene Nanoribbons. *Nanomaterials* **13**(14), 2066 (2023). DOI: 10.3390/nano13142066
- Ren, C.-Y., **Paudel, R. K.**, & Chang, Y.-C. Density Functional Theory for Buckyballs within Symmetrized Icosahedral Basis. *Nanomaterials* **13**(13), 1912 (2023). DOI: 10.3390/nano13131912

Manuscripts in Preparation

- **Paudel, R. K.**, et al. Sublattice-resolved molecule–graphene hybridization and its signature in quantum transport. In preparation (2026).

Conference Presentations

Oral Presentations

- Cha, S., Xu, Z., Ouyang, T., Yao, H., **Paudel, R. K.**, Taniguchi, T., Watanabe, K., Joe, A. Y., Chang, Y.-C., Gabor, N. M., & Lui, C. H. Hybridization-revealed interlayer 2p exciton state in bilayer MoS_2. *24th Workshop on First-Principles Computational Materials Physics*, Tainan, Taiwan (2026).
- **Paudel, R. K.**, Ren, C.-Y., & Chang, Y.-C. Semiempirical Pseudopotential Method for Transition-Metal Dichalcogenides. *CCP2025 – 36th IUPAP Conference on Computational Physics*, Oak Ridge, USA (2025, virtual).

- Ouyang, T., Liu, E., Cha, S., **Paudel, R. K.**, et al. Brightening Interlayer Excitons by Electric-Field-Driven Hole Transfer in Bilayer WSe_2. *23rd Workshop on First-Principles Computational Materials Physics*, Kaohsiung, Taiwan (2025).
- **Paudel, R. K.** Development of Semi-Empirical Pseudopotential Method for Two-Dimensional Materials. *20th Workshop on First-Principles Computational Materials Physics*, Kaohsiung, Taiwan (2022).

Poster Presentations

- Cha, S., Xu, Z., Ouyang, T., Yao, H., **Paudel, R. K.**, et al. Hybridization-revealed interlayer 2p exciton state in bilayer MoS_2. *26th Asian Workshop on First-Principles Electronic Structure Calculations (ASIAN-26)*, RIKEN, Japan (2025).
- **Paudel, R. K.** Semi-Empirical Pseudopotential Methods for Low-Dimensional Materials. *MRS Spring Meeting*, Seattle, WA, USA (2024).
- **Paudel, R. K.**, Ren, C.-Y., & Chang, Y.-C. Semi-Empirical Pseudopotential Method for Graphene and Armchair Graphene Nanoribbons. *CCP2023 – 34th IUPAP Conference on Computational Physics*, Kobe, Japan (2023).

Professional Affiliations

- Member, Taiwan Physical Society (TPS)
- Member, American Physical Society (APS)
- Life Member, Nepal Physical Society (NPS)

References

Prof. Yia-Chung Chang
Distinguished Research Fellow Emeritus
Research Center for Applied Sciences (RCAS), Academia Sinica, Taipei, Taiwan
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